

# Exam Formula Sheet

## Epi 202: Probability

$$\begin{aligned}\text{Var}(\tilde{a} \cdot \tilde{X}) &= \text{Var}\left(\sum_{i=1}^n a_i X_i\right) \\ &= \tilde{a}^\top \text{Var}(\tilde{X}) \tilde{a} \\ &= \sum_{i=1}^n \sum_{j=1}^n a_i a_j \text{Cov}(X_i, X_j)\end{aligned}$$

$$\text{E}[Y] = \text{E}[\text{E}[Y | X]]$$

$$\text{E}[Y | Z] = \text{E}[\text{E}[Y | X, Z] | Z]$$

$$\text{Var}(Y) = \text{E}[\text{Var}(Y | X)] + \text{Var}(\text{E}[Y | X])$$

$$\text{Cov}(Y, Z) = \text{E}[\text{Cov}(Y, Z | X)] + \text{Cov}(\text{E}[Y | X], \text{E}[Z | X])$$

## Epi 203: Statistical inference

$$\mathcal{L}(\theta) \stackrel{\text{def}}{=} \text{p}(\tilde{X} = \tilde{x} | \Theta = \theta)$$

$$\ell \stackrel{\text{def}}{=} \log\{\mathcal{L}(\tilde{x} | \theta)\}$$

$$\ell' \stackrel{\text{def}}{=} \frac{\partial}{\partial \theta} \ell(\tilde{x} | \theta)$$

$$\ell'' \stackrel{\text{def}}{=} \frac{\partial}{\partial \tilde{\theta}} \frac{\partial}{\partial \tilde{\theta}^\top} \ell(\tilde{x} | \tilde{\theta})$$

$$\ell''_{ij} = \frac{\partial}{\partial \theta_i} \frac{\partial}{\partial \theta_j} \ell(\tilde{X} = \tilde{x} | \tilde{\theta})$$

$$I \stackrel{\text{def}}{=} -\ell''(\tilde{x} | \tilde{\theta})$$

$$\mathcal{I} \stackrel{\text{def}}{=} \text{E}[I(\tilde{x} | \theta)]$$

$$\hat{\theta}_{ML} \sim \text{N}(\theta, [\mathcal{I}(\tilde{\theta})]^{-1})$$

For one parameter  $\theta_k$ :

$$\text{CI}_{1-\alpha}(\theta_k) = [\hat{\theta}_k \pm z_{1-\frac{\alpha}{2}} \widehat{\text{SE}}(\hat{\theta}_k)]$$

$$Z_k \stackrel{\text{def}}{=} \frac{\hat{\theta}_k - \theta_{k,0}}{\widehat{\text{SE}}(\hat{\theta}_k)} \sim \text{N}(0, 1) \quad \text{under } H_0 : \theta_k = \theta_{k,0} \quad z_k = \text{observed } Z_k$$

$$\begin{aligned}p\text{-value} &= 2 \text{Pr}(|Z| \geq |z_k|), \quad Z \sim \text{N}(0, 1) \\ &= 2(1 - \Phi(|z_k|))\end{aligned}$$

## Sta 108: Linear regression

$$t_k \stackrel{\text{def}}{=} \frac{\hat{\beta}_k - \beta_{k,0}}{\widehat{\text{SE}}(\hat{\beta}_k)} \sim t_{n-p} \quad \text{under } H_0 : \beta_k = \beta_{k,0} \quad t_k^{\text{obs}} = \text{observed } t_k$$

$$\text{CI}_{1-\alpha}(\beta_k) = \left[ \hat{\beta}_k \pm t_{n-p} \left( 1 - \frac{\alpha}{2} \right) \widehat{\text{SE}}(\hat{\beta}_k) \right]$$

Let  $T_{n-p} \sim t_{n-p}$ .

$$\begin{aligned} p\text{-value} &= 2 \Pr(|T_{n-p}| \geq |t_k^{\text{obs}}|) \\ &= 2(1 - F_{t_{n-p}}(|t_k^{\text{obs}}|)) \end{aligned}$$

$$\text{CI}_{1-\alpha}(\mu(\tilde{x}^*)) = \left[ \hat{\mu}(\tilde{x}^*) \pm t_{n-p} \left( 1 - \frac{\alpha}{2} \right) \widehat{\text{SE}}(\hat{\mu}(\tilde{x}^*)) \right]$$

$Y^*$  denotes a new observation (not in the training data), with corresponding covariate pattern  $\tilde{x}^*$ . Let  $\hat{Y}^* \stackrel{\text{def}}{=} \hat{\mu}(\tilde{x}^*)$ .

$$\text{PI}_{1-\alpha}(Y^*|\tilde{x}^*) = \left[ \hat{Y}^* \pm t_{n-p} \left( 1 - \frac{\alpha}{2} \right) \widehat{\text{SE}}(Y^* - \hat{Y}^*) \right]$$

$$\widehat{\text{SE}}(Y^* - \hat{Y}^*) = \hat{\sigma} \sqrt{1 + (\tilde{x}^*)^\top (\mathbf{X}'\mathbf{X})^{-1} \tilde{x}^*}$$

$$\text{Var}(Y^* - \hat{Y}^*) = \sigma^2 (1 + (\tilde{x}^*)^\top (\mathbf{X}'\mathbf{X})^{-1} \tilde{x}^*)$$

Let  $\hat{\Sigma} \stackrel{\text{def}}{=} \widehat{\text{Var}}(\hat{\beta}) = \hat{\sigma}^2 (\mathbf{X}'\mathbf{X})^{-1}$ .

$$\widehat{\text{Var}}(\hat{\mu}(\tilde{x})) = \tilde{x}^\top \hat{\Sigma} \tilde{x}$$

Let  $\Delta\mu(\tilde{x}, \tilde{x}^*) = \mu(\tilde{x}) - \mu(\tilde{x}^*)$ , and let  $\Delta\tilde{x} = \tilde{x} - \tilde{x}^*$ ; then:

$$\widehat{\text{Var}}(\widehat{\Delta\mu}(\tilde{x}, \tilde{x}^*)) = \Delta\tilde{x}^\top \hat{\Sigma} \Delta\tilde{x}$$

## Epi 204: Generalized linear models

Generalized linear models have three components:

1. The **outcome distribution** family:  $p(Y|\mu(\tilde{x}))$
2. The **link function**:  $g(\mu(\tilde{x})) = \eta(\tilde{x})$
3. The **linear component**:  $\eta(\tilde{x}) = \tilde{x} \cdot \beta$

$$\theta_\omega(\tilde{x}, \tilde{x}^*) = \exp\{(\Delta\tilde{x}) \cdot \tilde{\beta}\}$$

Estimates of odds ratios from 2x2 contingency tables

$$\hat{\theta} = \frac{ad}{bc}$$

Summary of logistic regression definitions and results

Odds and log-odds

These association-measure results are covered in Measures of Association for Binary Outcomes<sup>1</sup>.

Odds (def<sup>2</sup>)

$$\omega \stackrel{\text{def}}{=} \frac{\Pr(A)}{\Pr(\neg A)}$$

Conditional odds (def<sup>3</sup>)

$$\omega(A|B) \stackrel{\text{def}}{=} \frac{\Pr(A|B)}{\Pr(\neg A|B)}$$

Odds function (def<sup>4</sup>)

$$\text{odds}\{\pi\} \stackrel{\text{def}}{=} \frac{\pi}{1 - \pi}$$

<sup>1</sup>[binary-outcome-associations.qmd](#)

<sup>2</sup>[binary-outcome-associations.qmd#def-odds](#)

<sup>3</sup>[binary-outcome-associations.qmd#def-c-odds](#)

<sup>4</sup>[binary-outcome-associations.qmd#def-odds-fn](#)

**Probability to odds** (thm<sup>5</sup>, cor<sup>6</sup>)

$$\omega = \frac{\pi}{1 - \pi} = \text{odds}\{\pi\}$$

**Simplified odds expressions; odds of a non-event** (thm<sup>7</sup>, cor<sup>8</sup>)

$$\text{odds}\{\pi\} = \frac{1}{\pi^{-1} - 1} = (\pi^{-1} - 1)^{-1}, \quad \omega(\neg A) = \frac{1 - \pi}{\pi} = \pi^{-1} - 1$$

**Odds ratio** (def<sup>9</sup>)

$$\theta(\omega_1, \omega_2) \stackrel{\text{def}}{=} \frac{\omega_1}{\omega_2}$$

**OR as ratio of probability ratios** (thm<sup>10</sup>)

$$\theta(\omega_1, \omega_2) = \frac{\omega_1}{\omega_2} = \frac{\left(\frac{\pi_1}{1 - \pi_1}\right)}{\left(\frac{\pi_2}{1 - \pi_2}\right)}$$

**Odds ratios are reversible, marginally and conditionally** (thm<sup>11</sup>, thm<sup>12</sup>)

$$\theta_\omega(A|B) = \theta_\omega(B|A), \quad \theta_\omega(A|B, C) = \theta_\omega(B|A, C)$$

**Inverse-odds and probability recovery**

**Odds to probability** (thm<sup>13</sup>)

$$\pi = \frac{\omega}{1 + \omega}$$

**Inverse-odds function** (def<sup>14</sup>; recovers probability by cor<sup>15</sup>)

$$\text{invodds}\{\omega\} \stackrel{\text{def}}{=} \frac{\omega}{1 + \omega}, \quad \pi = \text{invodds}\{\omega\}$$

**Simplified inverse-odds** (lem<sup>16</sup>)

$$\text{invodds}\{\omega\} = \frac{1}{1 + \omega^{-1}} = (1 + \omega^{-1})^{-1}$$

**One minus inverse-odds, and its complement** (lem<sup>17</sup>, cor<sup>18</sup>)

$$1 - \pi = \frac{1}{1 + \omega}, \quad 1 + \omega = \frac{1}{1 - \pi}$$

**Log-odds (logit) and expit**

**Log-odds, and log-odds from probability** (def<sup>19</sup>, thm<sup>20</sup>)

$$\eta \stackrel{\text{def}}{=} \log\{\omega\}, \quad \eta = \log\left\{\frac{\pi}{1 - \pi}\right\}$$

**Logit function, and expanded form** (def<sup>21</sup>, thm<sup>22</sup>)

$$\text{logit}(\pi) \stackrel{\text{def}}{=} \log\{\text{odds}\{\pi\}\} = \log\left\{\frac{\pi}{1 - \pi}\right\}$$

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<sup>5</sup>[binary-outcome-associations.qmd#thm-prob-to-odds](#)

<sup>6</sup>[binary-outcome-associations.qmd#cor-oddsf-to-odds](#)

<sup>7</sup>[binary-outcome-associations.qmd#thm-odds-simplified](#)

<sup>8</sup>[binary-outcome-associations.qmd#cor-odds-neg](#)

<sup>9</sup>[binary-outcome-associations.qmd#def-OR](#)

<sup>10</sup>[binary-outcome-associations.qmd#thm-or-ratio-ratio](#)

<sup>11</sup>[binary-outcome-associations.qmd#thm-or-swap](#)

<sup>12</sup>[binary-outcome-associations.qmd#thm-conditional-OR-swap](#)

<sup>13</sup>[binary-outcome-associations.qmd#thm-odds-to-prob](#)

<sup>14</sup>[binary-outcome-associations.qmd#def-inv-odds](#)

<sup>15</sup>[binary-outcome-associations.qmd#cor-invodds-pi](#)

<sup>16</sup>[binary-outcome-associations.qmd#lem-invodds-simplified](#)

<sup>17</sup>[binary-outcome-associations.qmd#lem-one-minus-odds-inv](#)

<sup>18</sup>[binary-outcome-associations.qmd#cor-inverse-odds-nonevent](#)

<sup>19</sup>[binary-outcome-associations.qmd#def-logodds](#)

<sup>20</sup>[binary-outcome-associations.qmd#thm-logodds-pi](#)

<sup>21</sup>[binary-outcome-associations.qmd#def-logit-fn](#)

<sup>22</sup>[binary-outcome-associations.qmd#thm-logit-function](#)

**Log-odds equals logit** (cor<sup>23</sup>)

$$\eta = \text{logit}\{\pi\}$$

**Odds and probability from log-odds** (lem<sup>24</sup>, thm<sup>25</sup>)

$$\omega = \exp\{\eta\}, \quad \pi = \frac{\exp\{\eta\}}{1 + \exp\{\eta\}}$$

**Expit / inverse-logit function, and equivalent expressions** (def<sup>26</sup>, thm<sup>27</sup>)

$$\text{expit}(\eta) \stackrel{\text{def}}{=} \text{invodds}\{\exp\{\eta\}\} = \frac{\exp\{\eta\}}{1 + \exp\{\eta\}} = (1 + \exp\{-\eta\})^{-1}$$

**Probability as expit** (thm<sup>28</sup>)

$$\pi = \text{expit}\{\eta\} \tag{1}$$

**Logit and expit are inverses** (thm<sup>29</sup>)

$$\text{logit}\{\text{expit}\{\eta\}\} = \eta \quad \text{expit}\{\text{logit}\{\pi\}\} = \pi$$

$$\left[ \pi \stackrel{\text{def}}{=} \Pr(Y = 1 | \tilde{X} = \tilde{x}) \right] \xrightleftharpoons[\underbrace{\frac{\omega}{1+\omega}}_{\text{expit}(\eta)}]{\overbrace{\frac{\pi}{1-\pi}}^{\text{logit}(\pi)}} \left[ \omega \stackrel{\text{def}}{=} \text{odds}\{Y = 1 | \tilde{X} = \tilde{x}\} \right] \xrightleftharpoons[\underbrace{\exp\{\eta\}}_{\text{expit}(\eta)}]{\overbrace{\log\{\omega\}}^{\text{logit}(\omega)}} \left[ \eta(\tilde{x}) \stackrel{\text{def}}{=} \eta(Y = 1 | \tilde{X} = \tilde{x}) \right]$$

Figure 1: Diagram of logistic regression link and inverse link functions

**Rare events**

**Odds minus probability** (thm<sup>30</sup>)

$$\omega - \pi = \frac{\pi^2}{1 - \pi}, \quad \text{where } \omega = \frac{\pi}{1 - \pi}$$

**OR as RR times a correction factor** (thm<sup>31</sup>)

$$\theta(\omega_1, \omega_2) = \rho(\pi_1, \pi_2) \cdot \frac{1 - \pi_2}{1 - \pi_1}$$

When  $\pi_1$  and  $\pi_2$  are both small, the correction factor is close to 1, so the odds ratio is close to the risk ratio.

**Derivatives**

|                 | $\pi$                                                 | $\omega$                                        | $\eta$                                    | $\tilde{x}$                            | $\tilde{\beta}$                        |
|-----------------|-------------------------------------------------------|-------------------------------------------------|-------------------------------------------|----------------------------------------|----------------------------------------|
| $\pi$           | 1                                                     | $(1 + \omega)^2$                                | $\frac{(1 + \omega)^2}{\omega}$           | undef                                  | undef                                  |
| $\omega$        | $(1 - \pi)^2$                                         | 1                                               | $\frac{1}{\omega}$                        | undef                                  | undef                                  |
| $\eta$          | $\pi(1 - \pi)$                                        | $\omega$                                        | 1                                         | undef                                  | undef                                  |
| $\tilde{x}$     | $\underbrace{\tilde{\beta}\pi(1 - \pi)}_{p \times 1}$ | $\underbrace{\tilde{\beta}\omega}_{p \times 1}$ | $\underbrace{\tilde{\beta}}_{p \times 1}$ | $\underbrace{\mathbb{I}}_{p \times p}$ | $\mathbf{0}_{p \times p}$              |
| $\tilde{\beta}$ | $\underbrace{\tilde{x}\pi(1 - \pi)}_{p \times 1}$     | $\underbrace{\tilde{x}\omega}_{p \times 1}$     | $\underbrace{\tilde{x}}_{p \times 1}$     | $\mathbf{0}_{p \times p}$              | $\underbrace{\mathbb{I}}_{p \times p}$ |

<sup>23</sup>binary-outcome-associations.qmd#cor-logodds-logit

<sup>24</sup>binary-outcome-associations.qmd#lem-odds-from-logodds

<sup>25</sup>binary-outcome-associations.qmd#thm-prob-from-logodds

<sup>26</sup>binary-outcome-associations.qmd#def-expit

<sup>27</sup>binary-outcome-associations.qmd#thm-expit-expressions

<sup>28</sup>binary-outcome-associations.qmd#thm-expit-prob-logodds

<sup>29</sup>binary-outcome-associations.qmd#thm-logit-expit

<sup>30</sup>binary-outcome-associations.qmd#thm-odds-minus-probs

<sup>31</sup>binary-outcome-associations.qmd#thm-rare-disease-or-rr

Column labels indicate the numerators of the derivatives; row labels indicate the denominators (dimensions and details in tbl<sup>32</sup>).

**Derivative of expit** (cor<sup>33</sup>)

$$\text{expit}'\{\eta\} = \text{expit}\{\eta\}(1 - \text{expit}\{\eta\})$$

**Logistic regression model**

**Logistic regression** (def<sup>34</sup>)

$$Y_i | \tilde{X}_i \sim_{\perp} \text{Ber}(\pi(\tilde{X}_i)), \quad \text{logit}\{\pi(\tilde{x})\} = \tilde{x}^\top \tilde{\beta}$$

Throughout this summary,  $\tilde{\beta} = (\beta_0, \beta_1, \dots, \beta_p)$  is the  $(p+1) \times 1$  coefficient vector (including the intercept),  $\mathbf{X}$  is the  $n \times (p+1)$  design matrix whose  $i^{\text{th}}$  row is  $\tilde{x}_i^\top$ , and  $\tilde{\varepsilon}$  is the  $n \times 1$  error vector with components  $\varepsilon_i \stackrel{\text{def}}{=} y_i - \pi_i$ .

**Log-likelihood and score function**

**Log-likelihood component** (lem<sup>35</sup>)

$$\ell_i(\pi_i) = y_i \eta_i - \log\{1 + \omega_i\}$$

**Score function as sum of components** (lem<sup>36</sup>, thm<sup>37</sup>)

$$\tilde{\ell}'(\tilde{\beta}) = \sum_{i=1}^n \tilde{\ell}'_i(\tilde{\beta}), \quad \ell'_i(\tilde{\beta}) = \tilde{x}_i \varepsilon_i$$

**Score function** (thm<sup>38</sup>)

$$\tilde{\ell}'(\tilde{\beta}) = \sum_{i=1}^n \tilde{x}_i \varepsilon_i = \underbrace{\mathbf{X}^\top}_{(p+1) \times n} \underbrace{\tilde{\varepsilon}}_{n \times 1}$$

**Maximum likelihood estimation**

**Estimating equation** (eq<sup>39</sup>) The MLE  $\hat{\tilde{\beta}}$  solves the score equation:

$$\underbrace{\tilde{\ell}'(\hat{\tilde{\beta}})}_{(p+1) \times 1} = \sum_{i=1}^n \underbrace{\tilde{x}_i}_{(p+1) \times 1} \underbrace{\left( y_i - \text{expit}\left\{ \tilde{x}_i^\top \hat{\tilde{\beta}} \right\} \right)}_{1 \times 1} = \mathbf{0}_{(p+1) \times 1}$$

This estimating equation has no closed-form solution in general;  $\hat{\tilde{\beta}}$  must be computed by numerical iteration.

**Hessian** (eq<sup>40</sup>)

$$\underbrace{\ell''(\tilde{\beta})}_{(p+1) \times (p+1)} = - \underbrace{\mathbf{X}^\top}_{(p+1) \times n} \underbrace{\mathbf{D}}_{n \times n} \underbrace{\mathbf{X}}_{n \times (p+1)}, \quad \mathbf{D} \stackrel{\text{def}}{=} \text{diag}(\text{Var}(Y_i | \tilde{X}_i = \tilde{x}_i)) = \text{diag}(\pi_i(1 - \pi_i))$$

**Concavity** (rem<sup>41</sup>): the Hessian is negative semi-definite for every  $\tilde{\beta}$  (negative definite when  $\mathbf{X}$  has full column rank), so the log-likelihood is concave and every solution of the score equation is a global maximum.

**Newton-Raphson update** (Newton-Raphson method<sup>42</sup>)

$$\underbrace{\hat{\tilde{\beta}}^*}_{(p+1) \times 1} \leftarrow \underbrace{\hat{\tilde{\beta}}}_{(p+1) \times 1} - \underbrace{\left( \ell''(\tilde{y}; \hat{\tilde{\beta}}^*) \right)^{-1}}_{(p+1) \times (p+1)} \underbrace{\ell'(\tilde{y}; \hat{\tilde{\beta}}^*)}_{(p+1) \times 1}$$

**One-sample MLE for odds** (thm<sup>43</sup>)

$$\hat{\omega} = \frac{x}{n - x}$$

<sup>32</sup>logistic-regression.qmd#tbl-logistic-deriv-matrix

<sup>33</sup>logistic-regression.qmd#cor-deriv-expit

<sup>34</sup>logistic-regression.qmd#def-logistic-regression

<sup>35</sup>logistic-regression.qmd#lem-logistic-loglik-component

<sup>36</sup>logistic-regression.qmd#lem-score-logistic

<sup>37</sup>logistic-regression.qmd#thm-logistic-score-comp

<sup>38</sup>logistic-regression.qmd#thm-logistic-score-fn

<sup>39</sup>logistic-regression.qmd#eq-score-logistic

<sup>40</sup>logistic-regression.qmd#eq-logistic-hess

<sup>41</sup>logistic-regression.qmd#rem-hessian-nsd

<sup>42</sup>intro-MLEs.qmd#sec-newton-raphson

<sup>43</sup>binary-outcome-associations.qmd#thm-est-odds

## Wald tests and confidence intervals for coefficients

Wald test statistic for  $H_0 : \beta_k = \beta_{k,0}$  (CLT for MLEs<sup>44</sup>)

$$z_k = \frac{\hat{\beta}_k - \beta_{k,0}}{\widehat{\text{SE}}(\hat{\beta}_k)}, \quad z_k \sim N(0, 1) \text{ under } H_0$$

95% confidence intervals for  $\beta_k$  and  $e^{\beta_k}$  (MLE invariance)

$$\hat{\beta}_k \pm 1.96 \cdot \widehat{\text{SE}}(\hat{\beta}_k), \quad e^{\beta_k} \in \left( e^{\hat{\beta}_k - 1.96 \cdot \widehat{\text{SE}}(\hat{\beta}_k)}, e^{\hat{\beta}_k + 1.96 \cdot \widehat{\text{SE}}(\hat{\beta}_k)} \right)$$

For  $k \neq 0$ ,  $e^{\beta_k}$  is an odds ratio; for  $k = 0$ ,  $e^{\beta_0}$  is the baseline odds.

## Odds ratios in logistic regression

General OR formula (lem<sup>45</sup>)

$$\theta_\omega(\tilde{x}, \tilde{x}^*) = \exp\{\eta(\tilde{x}) - \eta(\tilde{x}^*)\}$$

Difference in log-odds; OR in terms of  $\Delta\eta$  (def<sup>46</sup>, cor<sup>47</sup>)

$$\Delta\eta \stackrel{\text{def}}{=} \eta(\tilde{x}) - \eta(\tilde{x}^*), \quad \theta_\omega(\tilde{x}, \tilde{x}^*) = \exp\{\Delta\eta\}$$

$\Delta\eta$  from covariates (lem<sup>48</sup>)

$$\Delta\eta = (\tilde{x} - \tilde{x}^*) \cdot \tilde{\beta}$$

Difference in covariate patterns;  $\Delta\eta$  from  $\Delta\tilde{x}$  (def<sup>49</sup>, cor<sup>50</sup>)

$$\Delta\tilde{x} \stackrel{\text{def}}{=} \tilde{x} - \tilde{x}^*, \quad \Delta\eta = \Delta\tilde{x} \cdot \tilde{\beta}$$

OR in terms of  $\Delta\tilde{x}$  (thm<sup>51</sup>)

$$\theta_\omega(\tilde{x}, \tilde{x}^*) = \exp\{(\Delta\tilde{x}) \cdot \tilde{\beta}\}$$

Log OR equals  $\Delta\eta$  (cor<sup>52</sup>)

$$\log\{\theta_\omega(\tilde{x}, \tilde{x}^*)\} = \Delta\eta$$

## Inference for log-odds and odds ratios

Estimated SE of log-odds (thm<sup>53</sup>)

$$\widehat{\text{Var}}(\hat{\eta}(\tilde{x})) = \underbrace{\tilde{x}^\top}_{1 \times (p+1)} \underbrace{\hat{\Sigma}}_{(p+1) \times (p+1)} \underbrace{\tilde{x}}_{(p+1) \times 1}, \quad \widehat{\text{SE}}(\hat{\eta}(\tilde{x})) = \sqrt{\tilde{x}^\top \hat{\Sigma} \tilde{x}}$$

Estimated SE of  $\Delta\hat{\eta}$  (thm<sup>54</sup>)

$$\widehat{\text{Var}}(\Delta\hat{\eta}) = \underbrace{\Delta\tilde{x}^\top}_{1 \times (p+1)} \underbrace{\hat{\Sigma}}_{(p+1) \times (p+1)} \underbrace{(\Delta\tilde{x})}_{(p+1) \times 1}, \quad \widehat{\text{SE}}(\Delta\hat{\eta}) = \sqrt{\Delta\tilde{x}^\top \hat{\Sigma} (\Delta\tilde{x})}$$

CI for an odds ratio (eq<sup>55</sup>; details in sec<sup>56</sup>): exponentiate the endpoints of the CI for  $\Delta\eta$ :

$$\theta_\omega(\tilde{x}, \tilde{x}^*) \in (e^L, e^R), \quad (L, R) = \widehat{\Delta\eta} \mp 1.96 \cdot \widehat{\text{SE}}(\widehat{\Delta\eta})$$

<sup>44</sup>intro-MLEs.qmd#thm-dist-mle

<sup>45</sup>logistic-regression.qmd#lem-logistic-OR

<sup>46</sup>logistic-regression.qmd#def-difflogodds

<sup>47</sup>logistic-regression.qmd#cor-logistic-OR

<sup>48</sup>logistic-regression.qmd#lem-diff-logodds

<sup>49</sup>logistic-regression.qmd#def-diff-covs

<sup>50</sup>logistic-regression.qmd#cor-diff-logodds

<sup>51</sup>logistic-regression.qmd#thm-logistic-OR

<sup>52</sup>logistic-regression.qmd#cor-log-or

<sup>53</sup>logistic-regression.qmd#thm-est-se-logodds

<sup>54</sup>logistic-regression.qmd#thm-est-se-diff-logodds

<sup>55</sup>logistic-regression.qmd#eq-ci-OR-exp

<sup>56</sup>logistic-regression.qmd#sec-or-inference

## Odds ratios in study designs

**Disease and exposure odds ratios** (def<sup>57</sup>, def<sup>58</sup>)

$$\theta_{\omega}(D|E) \stackrel{\text{def}}{=} \frac{\omega(D|E)}{\omega(D|\neg E)}, \quad \theta_{\omega}(E|D) \stackrel{\text{def}}{=} \frac{\omega(E|D)}{\omega(E|\neg D)}$$

**Case-control studies** (section<sup>59</sup>): outcome-stratified sampling breaks estimation of risks, risk ratios, and the components of the disease odds ratio (def<sup>60</sup>), but the exposure odds ratio (def<sup>61</sup>) is estimable, and it equals the disease odds ratio by reversibility (thm<sup>62</sup>).

**2×2 table shortcut** (table<sup>63</sup>; cell counts  $a, b$  = exposed events and non-events,  $c, d$  = unexposed events and non-events):

$$\hat{\omega}(\text{Event}|\text{Exposed}) = \frac{a}{b}, \quad \hat{\omega}(\text{Event}|\neg \text{Exposed}) = \frac{c}{d}, \quad \theta_{\omega}(\text{Exposed}, \neg \text{Exposed}) = \frac{ad}{bc}$$

## Model fit and comparison

Details in sec<sup>64</sup>; here  $q$  is the number of distinct covariate patterns,  $p$  is the number of  $\beta$  parameters, and  $\ell_{\text{full}}$  is the saturated-model log-likelihood.

**Deviance test** (def<sup>65</sup>)

$$D \stackrel{\text{def}}{=} 2(\ell_{\text{full}} - \ell(\hat{\beta}; \mathbf{x})), \quad D \sim \chi^2(q - p)$$

**Hosmer-Lemeshow test** (def<sup>66</sup>;  $g$  groups by predicted probability)

$$X^2 \stackrel{\text{def}}{=} \sum_{c=1}^g \frac{(o_c - e_c)^2}{e_c} \sim \chi^2(g - 2)$$

**Pearson chi-squared GOF test** (sum of squared Pearson residuals, def<sup>67</sup>)

$$X^2 = \sum_{k=1}^q X_k^2 \sim \chi^2(q - p)$$

**AIC and BIC** [lower is better] (AIC<sup>68</sup> (Akaike 1974); BIC<sup>69</sup> (Schwarz 1978))

$$\text{AIC} = -2\ell(\hat{\theta}) + 2p, \quad \text{BIC} = -2\ell(\hat{\theta}) + p \log\{n\}$$

**Likelihood ratio test** (def<sup>70</sup>; for nested models with  $p_0 < p_1$  parameters — here  $\ell_1$  and  $\ell_0$  are the two fitted models' log-likelihoods, not the saturated model's)

$$\Lambda \stackrel{\text{def}}{=} 2(\ell_1(\hat{\theta}_1) - \ell_0(\hat{\theta}_0)), \quad \Lambda \sim \chi^2(p_1 - p_0) \text{ under } H_0$$

by Wilks' theorem<sup>71</sup>.

## Residual-based diagnostics

Residuals are computed per covariate pattern  $k$  (grouped data);  $h_k$  denotes the leverage of pattern  $k$  (def<sup>72</sup>).

**Response residual** (def<sup>73</sup>)

$$e_k \stackrel{\text{def}}{=} \bar{y}_k - \hat{\pi}(x_k)$$

**Pearson residual (and standardized version)** (def<sup>74</sup>, def<sup>75</sup>)

$$X_k \stackrel{\text{def}}{=} \frac{\bar{y}_k - \hat{\pi}_k}{\sqrt{\hat{\pi}_k(1 - \hat{\pi}_k)/n_k}}, \quad r_{P_k} \stackrel{\text{def}}{=} \frac{X_k}{\sqrt{1 - h_k}}$$

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<sup>57</sup>binary-outcome-associations.qmd#def-disease-OR

<sup>58</sup>binary-outcome-associations.qmd#def-exposure-OR

<sup>59</sup>binary-outcome-associations.qmd#sec-CC-studies

<sup>60</sup>binary-outcome-associations.qmd#def-disease-OR

<sup>61</sup>binary-outcome-associations.qmd#def-exposure-OR

<sup>62</sup>binary-outcome-associations.qmd#thm-or-swap

<sup>63</sup>binary-outcome-associations.qmd#tbl-2x2-generic

<sup>64</sup>logistic-regression.qmd#sec-gof

<sup>65</sup>logistic-regression.qmd#def-deviance-test

<sup>66</sup>logistic-regression.qmd#def-hosmer-lemeshow

<sup>67</sup>logistic-regression.qmd#def-pearson-residual

<sup>68</sup>Linear-models-overview.qmd#def-aic

<sup>69</sup>Linear-models-overview.qmd#def-bic

<sup>70</sup>logistic-regression.qmd#def-lrt

<sup>71</sup>intro-MLEs.qmd#thm-wilks

<sup>72</sup>logistic-regression.qmd#def-leverage-glm

<sup>73</sup>logistic-regression.qmd#def-response-residual-logistic

<sup>74</sup>logistic-regression.qmd#def-pearson-residual

<sup>75</sup>logistic-regression.qmd#def-standardized-pearson-residual

**Deviance residual (and standardized version)** (def<sup>76</sup>)

$$d_k \stackrel{\text{def}}{=} \text{sign}(y_k - n_k \hat{\pi}_k) \left\{ \sqrt{2[\ell_{\text{full}}(x_k) - \ell(\hat{\beta}; x_k)]} \right\}, \quad r_{D_k} \stackrel{\text{def}}{=} \frac{d_k}{\sqrt{1 - h_k}}$$

**Cook's distance** (def<sup>77</sup>)

$$D_k \stackrel{\text{def}}{=} \frac{(r_{D_k})^2}{p} \cdot \frac{h_k}{1 - h_k}$$

**Leverage** (def<sup>78</sup>):  $h_k$  is the  $k$ -th diagonal element of the weighted hat matrix (with  $\mathbf{X}$  here the  $q \times p$  design matrix of covariate patterns):

$$\mathbf{H} \stackrel{\text{def}}{=} \underbrace{\mathbf{W}^{1/2}}_{q \times q} \underbrace{\mathbf{X}}_{q \times p} \left( \underbrace{\mathbf{X}^\top \mathbf{W} \mathbf{X}}_{p \times p} \right)^{-1} \underbrace{\mathbf{X}^\top}_{p \times q} \underbrace{\mathbf{W}^{1/2}}_{q \times q}, \quad \mathbf{W} \stackrel{\text{def}}{=} \text{diag}(n_k \hat{\pi}_k (1 - \hat{\pi}_k))_{q \times q}$$

**Comparing probabilities**

**Risk difference** (def<sup>79</sup>)

$$\delta(\pi_1, \pi_2) \stackrel{\text{def}}{=} \pi_1 - \pi_2$$

**Risk ratio** (def<sup>80</sup>)

$$\rho(\pi_1, \pi_2) \stackrel{\text{def}}{=} \frac{\pi_1}{\pi_2}$$

**Relative risk difference** (def<sup>81</sup>; equals RR minus 1 by thm<sup>82</sup>)

$$\xi(\pi_1, \pi_2) \stackrel{\text{def}}{=} \frac{\delta(\pi_1, \pi_2)}{\pi_2} = \rho(\pi_1, \pi_2) - 1$$

**Marginal risks from logistic regression**

**Conditional predicted risk** (def<sup>83</sup>)

$$\pi(a, z) \stackrel{\text{def}}{=} \Pr(Y = 1 \mid A = a, Z = z)$$

**Conditional RD and RR** (def<sup>84</sup>, def<sup>85</sup>)

$$\text{RD}(z) \stackrel{\text{def}}{=} \pi(1, z) - \pi(0, z), \quad \text{RR}(z) \stackrel{\text{def}}{=} \frac{\pi(1, z)}{\pi(0, z)}$$

**Observed marginal risk** (def<sup>86</sup>; standardization form by thm<sup>87</sup>)

$$\pi(a) \stackrel{\text{def}}{=} \Pr(Y = 1 \mid A = a) = \mathbb{E}[\pi(a, Z) \mid A = a]$$

**Potential mean and causal marginal risk** (def<sup>88</sup>, def<sup>89</sup>; identification form by thm<sup>90</sup>)

$$\mu_a \stackrel{\text{def}}{=} \mathbb{E}[Y^a], \quad \pi_a \stackrel{\text{def}}{=} \Pr(Y^a = 1) = \mathbb{E}[\pi(a, Z)]$$

**No confounding: observed = causal** (cor<sup>91</sup>; the two estimands are contrasted in rem<sup>92</sup>)

$$Z \perp\!\!\!\perp A \implies \pi(a) = \pi_a$$

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<sup>76</sup>logistic-regression.qmd#def-deviance-residual-logistic

<sup>77</sup>logistic-regression.qmd#def-cooks-distance-glm

<sup>78</sup>logistic-regression.qmd#def-leverage-glm

<sup>79</sup>binary-outcome-associations.qmd#def-RD

<sup>80</sup>binary-outcome-associations.qmd#def-RR

<sup>81</sup>binary-outcome-associations.qmd#def-RRD

<sup>82</sup>binary-outcome-associations.qmd#thm-rrd-vs-rr

<sup>83</sup>logistic-regression.qmd#def-cond-predicted-risk

<sup>84</sup>logistic-regression.qmd#def-cond-rd

<sup>85</sup>logistic-regression.qmd#def-cond-rrr

<sup>86</sup>logistic-regression.qmd#def-observed-marginal-risk

<sup>87</sup>logistic-regression.qmd#thm-observed-marginal-risk

<sup>88</sup>logistic-regression.qmd#def-potential-mean

<sup>89</sup>logistic-regression.qmd#def-causal-marginal-risk

<sup>90</sup>logistic-regression.qmd#thm-causal-marginal-risk

<sup>91</sup>logistic-regression.qmd#cor-observed-equals-causal

<sup>92</sup>logistic-regression.qmd#rem-observed-vs-causal-marginal-risk



**G-computation estimator** (def<sup>93</sup>; consistent for  $\pi_a$  by thm<sup>94</sup>)

$$\hat{\pi}_a \stackrel{\text{def}}{=} \frac{1}{n} \sum_{i=1}^n \hat{\pi}_{a,i}, \quad \hat{\pi}_{a,i} \stackrel{\text{def}}{=} \hat{\pi}(a, z_i)$$

**Predictive margins** (def<sup>95</sup>; full-sample form def<sup>96</sup> is consistent for  $\pi_a$  by cor<sup>97</sup>)

$$\widehat{\text{PM}}(a \mid S) \stackrel{\text{def}}{=} \frac{1}{|S|} \sum_{i \in S} \hat{\pi}_{a,i}, \quad \widehat{\text{PM}}(a) \stackrel{\text{def}}{=} \frac{1}{n} \sum_{i=1}^n \hat{\pi}_{a,i} = \hat{\pi}_a$$

**Exposed-subgroup predictive margin** (def<sup>98</sup>; consistent for  $\pi(a)$  by thm<sup>99</sup>; equals  $\bar{Y}_a$  exactly by lem<sup>100</sup>)

$$\hat{\pi}_a(a) \stackrel{\text{def}}{=} \frac{1}{n_a} \sum_{i: A_i=a} \hat{\pi}_{a,i} = \bar{Y}_a$$

**Observed marginal RD and RR** (def<sup>101</sup>, def<sup>102</sup>)

$$\text{OMRD} \stackrel{\text{def}}{=} \pi(1) - \pi(0), \quad \text{OMRR} \stackrel{\text{def}}{=} \frac{\pi(1)}{\pi(0)}$$

**Causal marginal RD and RR** (def<sup>103</sup>, def<sup>104</sup>)

$$\text{CMRD} \stackrel{\text{def}}{=} \pi_1 - \pi_0, \quad \text{CMRR} \stackrel{\text{def}}{=} \frac{\pi_1}{\pi_0}$$

**Bootstrap SE and CI for marginal contrasts** (def<sup>105</sup>; see Bootstrap Confidence Intervals<sup>106</sup> and exm<sup>107</sup>): resample the data with replacement  $B$  times; refit the logistic model and recompute the marginal contrast on each resample; the bootstrap SE is the standard deviation of the  $B$  estimates, and a 95% percentile CI uses their 2.5th and 97.5th percentiles.

**Collapsibility** (def<sup>108</sup>): an estimand  $\theta$  is collapsible with respect to  $Z$  if

$$\theta \stackrel{\text{def}}{=} \text{E}[\theta(Z)]$$

Risk differences are collapsible; risk ratios and odds ratios are not collapsible in general (thm<sup>109</sup>; counterexample in exm<sup>110</sup>).

## Survival analysis

### Probability distribution functions

Table 1: Probability distribution functions

| Name                                   | Symbols            | Definition                         |
|----------------------------------------|--------------------|------------------------------------|
| Probability density function (PDF)     | $f(t), p(t)$       | $p(T = t)$                         |
| Cumulative distribution function (CDF) | $F(t), P(t)$       | $P(T \leq t)$                      |
| Survival function                      | $S(t), \bar{F}(t)$ | $P(T > t)$                         |
| Hazard function                        | $\lambda(t), h(t)$ | $p(T = t \mid T \geq t)$           |
| Cumulative hazard function             | $\Lambda(t), H(t)$ | $\int_{u=-\infty}^t \lambda(u) du$ |
| Log-hazard function                    | $\eta(t)$          | $\log\{\lambda(t)\}$               |

<sup>93</sup>logistic-regression.qmd#def-g-computation

<sup>94</sup>logistic-regression.qmd#thm-g-computation-consistency

<sup>95</sup>logistic-regression.qmd#def-predictive-margins

<sup>96</sup>logistic-regression.qmd#def-full-sample-predictive-margin

<sup>97</sup>logistic-regression.qmd#cor-full-sample-pm

<sup>98</sup>logistic-regression.qmd#def-subgroup-predictive-margin

<sup>99</sup>logistic-regression.qmd#thm-subgroup-pm

<sup>100</sup>logistic-regression.qmd#lem-subgroup-pm-proportion

<sup>101</sup>logistic-regression.qmd#def-observed-marginal-rd

<sup>102</sup>logistic-regression.qmd#def-observed-marginal-rr

<sup>103</sup>logistic-regression.qmd#def-causal-marginal-rd

<sup>104</sup>logistic-regression.qmd#def-causal-marginal-rr

<sup>105</sup>logistic-regression.qmd#def-bootstrap-marginal-rd

<sup>106</sup>basic-statistical-methods.qmd#sec-bootstrap-ci

<sup>107</sup>logistic-regression.qmd#exm-wcgs-marginal-rd

<sup>108</sup>logistic-regression.qmd#def-collapsibility

<sup>109</sup>logistic-regression.qmd#thm-collapsibility

<sup>110</sup>logistic-regression.qmd#exm-collapsibility

### Diagram of survival distribution function relationships

$$\begin{aligned} f(t) &\xleftarrow[\frac{-S'(t)}{S(t)\lambda(t)}]{} S(t) \xleftarrow[\exp\{-\Lambda(t)\}]{} \Lambda(t) \xleftarrow[\int_{u=0}^t \lambda(u)du]{} \lambda(t) \xleftarrow[\exp\{\eta(t)\}]{} \eta(t) \\ f(t) &\xrightarrow[\int_{u=t}^{\infty} f(u)du]{} S(t) \xrightarrow[-\log S(t)]{} \Lambda(t) \xrightarrow[\Lambda'(t)]{} \lambda(t) \xrightarrow[\log\{\lambda(t)\}]{} \eta(t) \end{aligned}$$

### Survival likelihood contributions, assuming non-informative censoring

$$\begin{aligned} p(Y = y, D = d) &= [f_T(y)]^d [S_T(y)]^{1-d} \\ &= [\lambda_T(y)]^d [S_T(y)] \end{aligned}$$

### Nonparametric time-to-event distribution estimators

$$\begin{aligned} \hat{\lambda}_i &= \frac{d_i}{r_i} \\ \hat{\kappa}_i &= 1 - \hat{\lambda}_i = \frac{r_i - d_i}{r_i} \\ \hat{S}_{\text{KM}}(t) &\stackrel{\text{def}}{=} \prod_{\{i: t_i \leq t\}} \hat{\kappa}_i \\ \hat{\Lambda}_{\text{NA}}(t) &\stackrel{\text{def}}{=} \sum_{\{i: t_i \leq t\}} \hat{\lambda}_i \end{aligned}$$

### Proportional hazards model structure

| Formula                                                                                                                                                         | Description                     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| $\lambda(t   \tilde{x}) = \lambda_0(t) \cdot \theta_\lambda(\tilde{x})$                                                                                         | Proportional hazards assumption |
| $\Lambda(t   \tilde{x}) = \Lambda_0(t) \cdot \theta_\lambda(\tilde{x})$                                                                                         | Cumulative hazard factorization |
| $\eta(t   \tilde{x}) = \eta_0(t) + \Delta\eta(\tilde{x})$                                                                                                       | Log-hazard decomposition        |
| $\Delta\eta(\tilde{x}) = \tilde{x} \cdot \tilde{\beta} = \beta_1 x_1 + \dots + \beta_p x_p$                                                                     | Linear predictor                |
| $\theta_\lambda(\tilde{x}) = \exp\{\Delta\eta(\tilde{x})\}$                                                                                                     | Hazard multiplier               |
| $\theta_\lambda(t   \tilde{x} : \tilde{x}^*) \stackrel{\text{def}}{=} \frac{\lambda(t   \tilde{x})}{\lambda(t   \tilde{x}^*)}$                                  | Hazard ratio definition         |
| $\theta_\lambda(t   \tilde{x} : \tilde{x}^*) = \exp\{\Delta\eta(\tilde{x}) - \Delta\eta(\tilde{x}^*)\} = \exp\{(\tilde{x} - \tilde{x}^*) \cdot \tilde{\beta}\}$ | Hazard ratio formula            |

### Proportional hazards model partial likelihood formula:

**Definition 0.1** (Cox PH partial likelihood).

$$\begin{aligned} \mathcal{L}_i^*(\tilde{\beta}) &\stackrel{\text{def}}{=} \frac{\theta_\lambda(\tilde{x}_{(i)})}{\sum_{k \in R(t_i)} \theta_\lambda(\tilde{x}_k)} \\ \mathcal{L}^*(\tilde{\beta}) &\stackrel{\text{def}}{=} \prod_{\{i: d_i=1\}} \mathcal{L}_i^*(\tilde{\beta}) \end{aligned}$$

where, for the distinct event times  $t_1 < \dots < t_K$  of the data set:

- $(i)$  is the subject who fails at  $t_i$  (unique under the no-ties assumption);
- $R(t_i) \stackrel{\text{def}}{=} \{j : Y_j \geq t_i\}$  is the risk set at  $t_i$  (where  $Y_j = \min(T_j, C_j)$  is subject  $j$ 's observed follow-up time) — the subjects still under observation just before  $t_i$ ;
- $d_i$  is the number of events at  $t_i$  (so  $d_i = 1$  when there are no ties);
- $\theta_\lambda(\tilde{x}) \stackrel{\text{def}}{=} \exp\{\tilde{x}^\top \tilde{\beta}\}$  is the hazard ratio (risk score) for covariate pattern  $\tilde{x}$ .

### Proportional hazards model baseline cumulative hazard estimator:

**Definition 0.2** (Breslow estimator of the baseline cumulative hazard).

$$\hat{\Lambda}_0(t) \stackrel{\text{def}}{=} \sum_{t_i \leq t} \frac{d_i}{\sum_{k \in R(t_i)} \theta_\lambda(\tilde{x}_k)}$$

where, for the distinct event times  $t_1 < \dots < t_K$  of the data set:

- $d_i$  is the number of events at  $t_i$ ;
- $R(t_i) \stackrel{\text{def}}{=} \{j : Y_j \geq t_i\}$  is the risk set at  $t_i$  (where  $Y_j = \min(T_j, C_j)$  is subject  $j$ 's observed follow-up time) — the subjects still under observation just before  $t_i$ ;
- $\theta_\lambda(\tilde{x}) \stackrel{\text{def}}{=} \exp\{\tilde{x}^\top \tilde{\beta}\}$  is the hazard ratio (risk score) for covariate pattern  $\tilde{x}$ .

Exm

**Example 0.1** (Breslow estimator with two event times). Suppose there are two distinct event times  $t_1 = 2$  and  $t_2 = 4$  (no ties,  $d_1 = d_2 = 1$ ), with risk-set-weighted totals  $\sum_{k \in R(2)} \theta_\lambda(\tilde{x}_k) = 10$  and  $\sum_{k \in R(4)} \theta_\lambda(\tilde{x}_k) = 6$ . Then

$$\hat{\Lambda}_0(4) = \frac{1}{10} + \frac{1}{6} \approx 0.267.$$

Akaike, Hirotugu. 1974. “A New Look at the Statistical Model Identification.” *IEEE Transactions on Automatic Control* 19 (6): 716–23. <https://doi.org/10.1109/TAC.1974.1100705>.

Schwarz, Gideon. 1978. “Estimating the Dimension of a Model.” *The Annals of Statistics* 6 (2): 461–64. <https://doi.org/10.1214/aos/1176344136>.